

CHAPTER – 4

IDENTIFY FUNCTIONAL DEPENDANCY, ANALYZE REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

In the Zara Fashion E-Commerce Management database, it is important to ensure that the database is well-structured, efficient, and free from unnecessary duplication. In this section, each entity is analyzed by identifying its functional dependencies to understand how attributes are related to one another. Redundancy analysis is performed to verify that duplicate data is minimized across the database tables. Data anomalies, including insertion, update, and deletion anomalies, are also examined to ensure data consistency, integrity, and reliable database operations. This analysis helps in designing a normalized and efficient database for the Zara Fashion E-Commerce Management System.

1.CUSTOMER TABLE

Functional Dependency

Customer_ID → First_Name, Email, Phone_Number, Address

Redundancy Analysis

- Customer details may be repeated in multiple orders if stored in the Order table.
- Keeping customer information only in the Customer table avoids duplication.

Data Anomalies

- Insertion Anomaly: Cannot create an order without first creating a customer.
- Update Anomaly: Updating a customer's phone number in multiple tables may cause inconsistency.
- Deletion Anomaly: Deleting the last order should not delete customer details.

2. CATEGORY

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Category information should not be stored repeatedly with every product.

Data Anomalies

- Insertion Anomaly: A category can exist without products.
- Update Anomaly: Changing a category name in multiple product records causes inconsistency.
- Deletion Anomaly: Deleting the last product should not remove the category.

3.PRODUCT

Functional Dependency

Product_ID → Product_Name, Brand, Price, Size, Colour, Category_ID

Redundancy Analysis

Product details should not be duplicated in Cart or Order tables except where historical pricing is required.

Data Anomalies

- Insertion Anomaly: Product cannot reference a non-existing category.
- Update Anomaly: Updating price in multiple tables creates inconsistency.
- Deletion Anomaly: Removing a product may affect cart and order references.

4. CART

Functional Dependency

Cart_ID → Customer_ID, Product_ID, Quantity

Redundancy Analysis

Repeating the same product in a cart increases redundancy unless managed properly.

Data Anomalies

- Insertion Anomaly: Cannot add a cart item without an existing customer and product.
- Update Anomaly: Quantity changes must be reflected correctly.
- Deletion Anomaly: Removing a cart item should not delete customer or product information.

5. ORDER

Functional Dependency

Order_ID $\rightarrow$ Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

Customer information should not be repeated in every order record.

Data Anomalies

- Insertion Anomaly: Order cannot exist without a customer.
- Update Anomaly: Order status updates should occur only once.
- Deletion Anomaly: Deleting an order should not delete customer information.

6. ORDER_ITEM

Functional Dependency

Order_Item_ID $\rightarrow$ Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

Product details should not be copied into every order item except the purchase price.

Data Anomalies

- Insertion Anomaly: Cannot create an order item without an order and product.
- Update Anomaly: Changing ordered quantity must update only one record.
- Deletion Anomaly: Removing an order item should not delete the order or product.

7. PAYMENT

Functional Dependency

Payment_ID $\rightarrow$ Order_ID, Payment_Method, Payment_Date, Payment_Status, Amount

Redundancy Analysis

Payment information should be stored only once for each payment.

Data Anomalies

- Insertion Anomaly : Payment cannot exist without an order.

- Update Anomaly: Updating payment status in multiple places causes inconsistency.
- Deletion Anomaly: Deleting payment records should not delete order information.

8. CONCLUSION

The functional dependency, redundancy, and data anomaly analysis confirms that the Zara Fashion E-Commerce Management database is logically structured and efficient. It helps reduce duplicate data, maintain data consistency, and improve overall database performance. This analysis also supports a reliable and maintainable database design for the e-commerce system.